

电路理论 mooc 答案——第十三章

仅限上海交通大学电路理论学科营使用

解析

请勿用于其他用途

1.解:

$$T = 1\text{ms} \Rightarrow f = 1\text{kHz} \Rightarrow \omega = 2\pi \text{krad/s}, \text{排除 } AC$$

$$B.u(2.5 \times 10^{-4}) = 20 \cos\left(\frac{1}{2}\pi - \frac{1}{3}\pi\right) > 0, \text{排除}$$

选D



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2.解:略,参见T1

选C

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3.解:有效值相量

选A

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Remark:没给参考相量变换下,一般取 $\sqrt{2}\cos\omega t$ 对应 $1\angle 0$

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4.解:

电感电容阻抗均为纯虚数,直接排除AC

电感电压超前电流 90° ,排除D

选B



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5.解:

$$u(t) = 220\sqrt{2}\cos(750\pi + 78^\circ) \Rightarrow \dot{U} = 220\angle 78^\circ$$

$$i(t) = 10\sqrt{2}\sin(750\pi + 123^\circ) = 10\sqrt{2}\cos(750\pi + 33^\circ) \Rightarrow \dot{I} = 10\angle 33^\circ$$

$$Z = \frac{\dot{U}}{\dot{I}} = 22\angle 45^\circ \Omega$$

选B

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6.解:

(a) 对含受控源支路诺顿变换后, 受控电流源可直接等效为 $\frac{1}{10}Z_L = 2j \Omega$

$$Z_{ab} = 10 + \frac{1}{\frac{1}{-j5} + \frac{1}{4} + \frac{1}{j2}} = 11.6 + 1.97j \Omega$$

(b) Let $\dot{U}_{ab} = 100 \angle 0^\circ$, then $\dot{U}_R = \frac{1}{2}\dot{U}_{ab} = 50 \angle 0^\circ = \dot{U}_{ab} - \dot{U}_R = \dot{U}_C$

$$\dot{I}_{ab} = \dot{I}_C + \left(-\frac{1}{n}\right)(\dot{I}_C - \dot{I}_R) = \frac{1}{2}(\dot{I}_C + \dot{I}_R) = \frac{1}{2}(10 \angle 90^\circ + 5 \angle 0^\circ) = 2.5 + 5j A$$

$$Z_{ab} = \frac{\dot{U}_{ab}}{\dot{I}_{ab}} = 8 - 16j \Omega$$

选C

7.解:

$$\text{Let } \dot{U} = 100 \angle 0^\circ, \text{ then } \dot{I} = \frac{\dot{U}}{Z_1 // Z_2} = 28 + 4j A, \dot{I}_1 = \frac{\dot{U}}{Z_1} = 12 + 16j A, \dot{I}_2 = \frac{\dot{U}}{Z_2} = 16 - 12j A$$

选D

Remark: 交流电表显示的都是有效值

8.解:

先对受控源戴维南等效, 合并 $\dot{U}_s$ 和 $aR_2\dot{I}$, 再诺顿等效, 最终得到诺顿电路 $\dot{I}_s = \frac{\dot{U}_s}{R_1 + R_2} + \frac{aR_2}{R_1 + R_2}\dot{I}$, $Z = (R_1 + R_2) // R_3 // \left(\frac{1}{j\omega C}\right)$

Remark: $Z \neq Z_{eq}$, 因为 $\dot{I}_s$ 中的 $\frac{aR_2}{R_1 + R_2}\dot{I}$ 分量也必须等效为阻抗

$$\dot{I}_Z = \dot{I}_s - \dot{I} = \frac{\dot{U}_s}{R_1 + R_2} - \frac{R_1 + (1-a)R_2}{R_1 + R_2}\dot{I} \Rightarrow \dot{U} = \dot{I}_Z Z = \left[\frac{\dot{U}_s}{R_1 + R_2} - \frac{R_1 + (1-a)R_2}{R_1 + R_2}\dot{I} \right] Z = \frac{Z}{R_1 + R_2}\dot{U}_s + \left[\frac{R_1 + (1-a)R_2}{R_1 + R_2} Z \right] (-\dot{I})$$

$$\therefore Z_{eq} = \frac{R_1 + (1-a)R_2}{R_1 + R_2} Z = \frac{R_1 + (1-a)R_2}{R_1 + R_2} \cdot \frac{1}{\frac{1}{R_1 + R_2} + \frac{1}{R_3} + j\omega C} = \frac{R_1 + (1-a)R_2}{R_1 + R_2} \cdot \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3 + j\omega CR_3(R_1 + R_2)} = \frac{R_3[R_1 + (1-a)R_2]}{R_1 + R_2 + R_3 + j\omega CR_3(R_1 + R_2)}$$

选B

9.解:

$$G_{12} = -[(j\omega L_1)^{-1} + (j\omega L_2)^{-1}], \text{ 排除BCD}$$

选A

10.解:

$$KVL: \dot{U} = j\omega L_1 \dot{I}_1 - j\omega M \dot{I} = -j\omega M \dot{I}_1 + j\omega L_2 \dot{I} + \dot{I}R$$

$$\text{消元, 寻找 } \dot{U}、\dot{I} \Rightarrow \frac{\dot{U} + j\omega M \dot{I}}{\dot{U} - j\omega L_2 \dot{I} - \dot{I}R} = \frac{j\omega L_1 \dot{I}_1}{-j\omega M \dot{I}_1} = -\frac{L_1}{M} \Rightarrow -M(\dot{U} + j\omega M \dot{I}) = L_1(\dot{U} - j\omega L_2 \dot{I} - \dot{I}R) \Rightarrow -j\omega M^2 \dot{I} = -j\omega L_1 L_2 \dot{I} \Rightarrow M = \sqrt{L_1 L_2}$$

选D